sometimes refer to as “new inflation,” became the basis of the kind of inflation we all talk about today.

3 Because we like to give particles and their associated fields names that end in “on.”

4 Each of these branes is designated in the official literature as an “end-of-the-world” brane, because it lies at the boundary of the space. This seems fitting.

5 The term “braneworld” specifically refers to models in which there are higher dimensions and our observable universe lives on a 3D brane within a larger space. It’s sort of a type of multiverse, but usually when people talk about a multiverse they’re referring to something different, like regions of a larger (3D) space where the laws of physics might be different, or even the Many Worlds interpretation of quantum mechanics, which is another matter entirely. Any construction that allows there to be more to reality than our observable cosmic volume is a kind of multiverse theory.

6 I am using “cyclic” and “bouncing” more or less interchangeably here, but a bouncing model doesn’t have to be cyclic, in the sense that there could be only one “bounce”—a transition from some past long-lived pre–Big Bang phase to our current universe, which then dies out on its own without producing a new universe afterward.

7 The second iteration of the Background Imaging of Cosmic Extragalactic Polarization experiment.

8 Technically, depending on the model, you could also get some tiny tiny level of primordial gravitational waves in an ekpyrotic universe during the slow contraction phase. But they’d be WAY too small to ever show up in observations.

9 The idea that there might be matter on the hidden brane has been discussed in the literature, but as far as I’m aware, detecting black hole collisions across the bulk has not. Perhaps it would require too many levels of speculation for a serious study. But I think it sounds like fun.